

WaterInnEU, WaCoDiS, MuDaK-WRM

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Hydrology Domain Working Group

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Geneva, Switzerland

WaterInnEU

- H2020 Coordination and Support Action
- March 2015 - February 2017
- Applying European market leadership to river basin networks and spreading of innovation on water ICT models, tools and data
- Two main developments
 - WaterInnEU Marketplace
 - RiBaSE pilot



WaterInnEU - Marketplace

- WaterInnEU Marketplace

Prototype



This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No 641821.

LoginRegisterNewsletterensearch portal



WaterInnEU Marketplace

Access to EU innovation for river basin management

HomeProductsOrganisationsMatchmaking & SupportE-learningWater Related Resources

Products & Services

Product of the Month

CORES-A1



The CORES-A1 is an unmanned robotised boat for ambient monitoring of coastal and inland water. The station is provided with appropriate navigation and measurement equipment, is managed remotely from the shore, and transmits the data in real time. The station provides cost effective sampling from the aquatic environment and direct... [Read more](#)

License / Copyright: Proprietary

You can also:

- Take a course to help you implement a specific product
- Join our community to add your own product or organisation, find support or join a discussion
- Go to the forum to see the latest debates on river basin management
- Subscribe for alerts to get informed about the latest changes depending on your requirements



WaterInnEU - RIBASE

- RIBASE: River BASin Standards intEroperability pilot
- Pilot for testing OGC Web Services for the integration of water-related information and models
- RiBaSE is designed for testing:
 - The adaptability of common spatial standards to water applications
 - The best suitable connection between these standards
 - The application of WaterML in a general geospatial framework

Acronym	Name
netCDF CF	Network Common Data Form
SAS draft	Sensor Alert Service
SOS 2.0	Sensor Observation Service
WaterML 2.0	Water Markup Language
WCS 2.0	Web Coverage Service
WFS 2.0	Web Feature Service
WPS 1.0	Web Processing Service

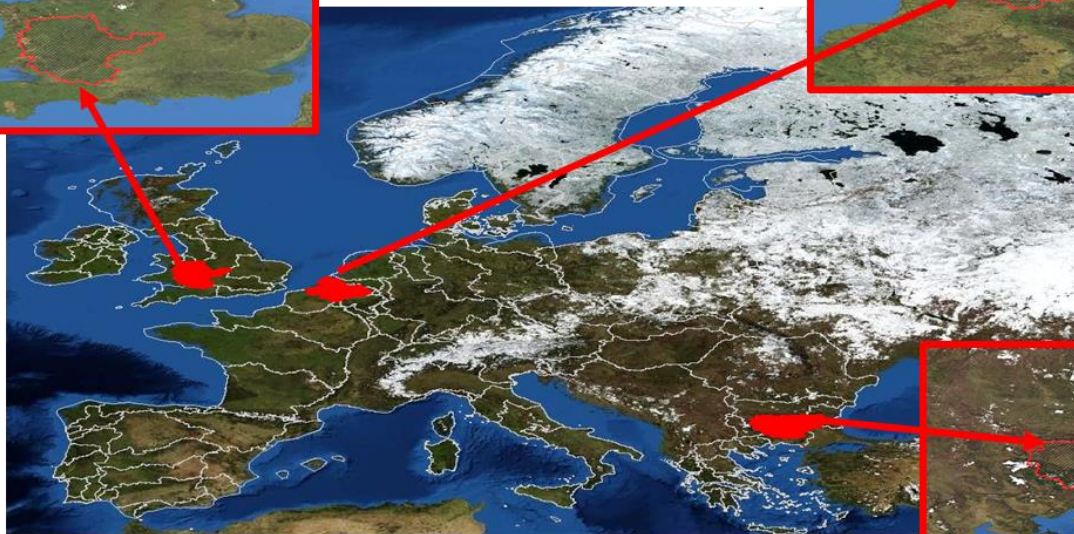
WaterInnEU - RIBASE



Severn



Scheldt

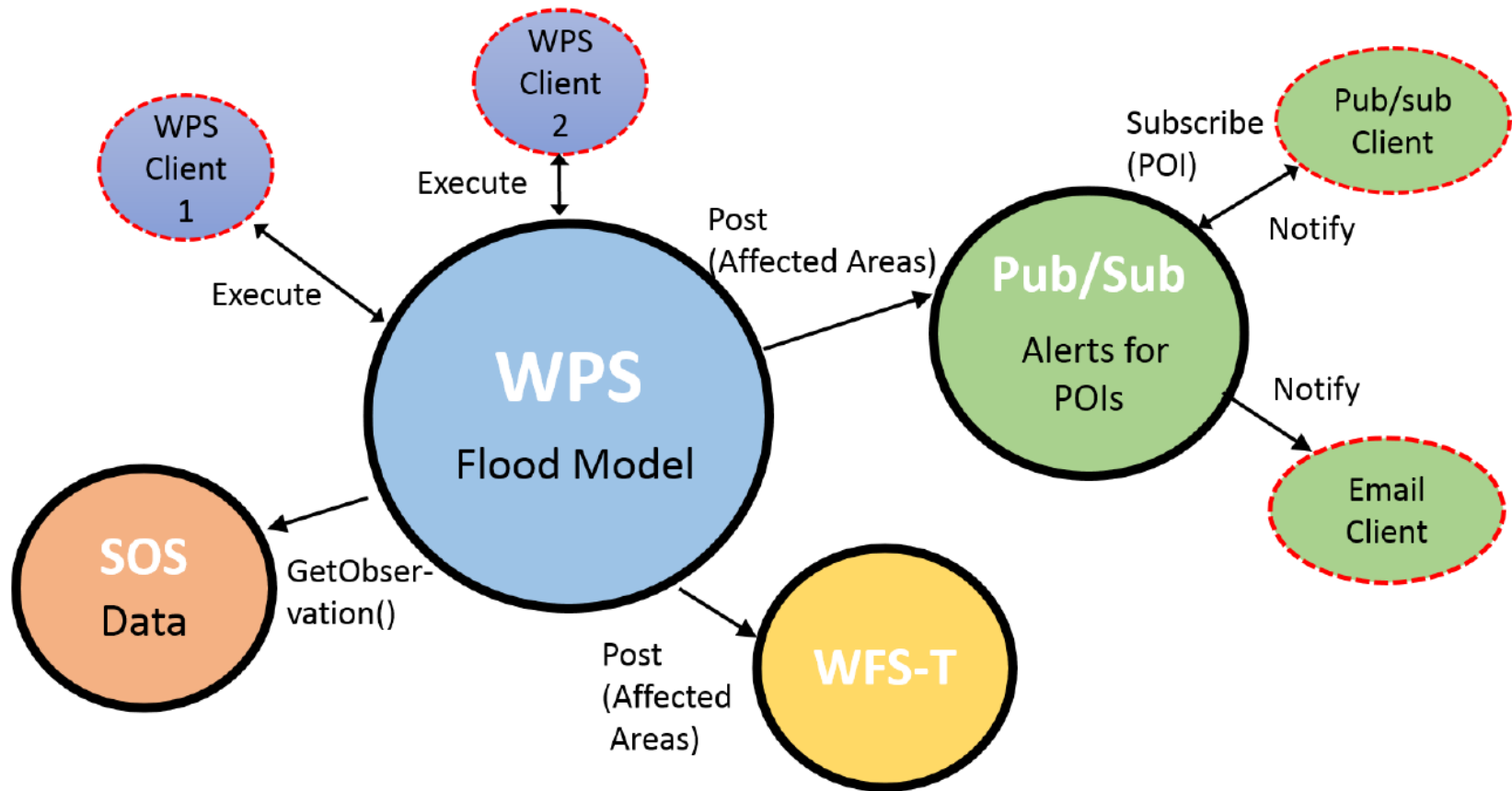


500 0 2000 km

Maritsa



WaterInnEU - RIBASE



WaterInnEU - RIBASE

- Flood scenario
 - Severn
 - SOS by National River Flow Archive (NRFA) hosted in the Centre for Ecology & Hydrology (CEH)

Centre for Ecology & Hydrology
NATURAL ENVIRONMENT RESEARCH COUNCIL

HOME OUR SCIENCE NEWS CENTRE DATA HOLDINGS PRODUCTS

You are here: CEH Web | National River Flow Archive | Search for data

National River Flow Archive: Search for gauging stations

1. Select search method
Search by gauged daily flow statistics

2. Select attribute to search on
Search by base flow index

3. Set search options
BFI > 0.7

4. Search peak flow or daily flow stations
Both

216 stations found.
Red dots show indicative station location only.
Click on the station number to access station and catchment information and, where available, download river flow data.

Stations displayed

No.	River	Location	Area (km ²)	GDF data period	GDF download
10001	Ythan	Ardlethen	448.1	1965-1982	N
10003	Ythan	Elton	523.0	1983-2013	N
11004	Urie	Pitcaple	196.0	1984-2013	N
26001	West Beck	Wansford Bridge	192.0	1953-1974	N
26002	Hull	Hempholme Lock	376.1	1961-2013	N
26003	Foston Beck	Foston Mill	57.2	1959-2013	Y
26004	Oypsey Race	Brndington	253.8	1971-1985	N
26005	Oypsey Race	Boynton	240.0	1981-2013	Y
26006	Elsworth Beck	1 mile Puffield	436.6	1980-2013	N

A map is provided to allow gauging stations to be located, and for contextual location information. Click below to turn this map off if download speeds for your internet connection cause this page to run too slowly.
☒ Show background map?

2D 3D Road Aerial Bird's eye Labels

250 miles

© 2014 Microsoft Corporation Image courtesy of NASA © 2010 Pearson

WaterInnEU - RIBASE

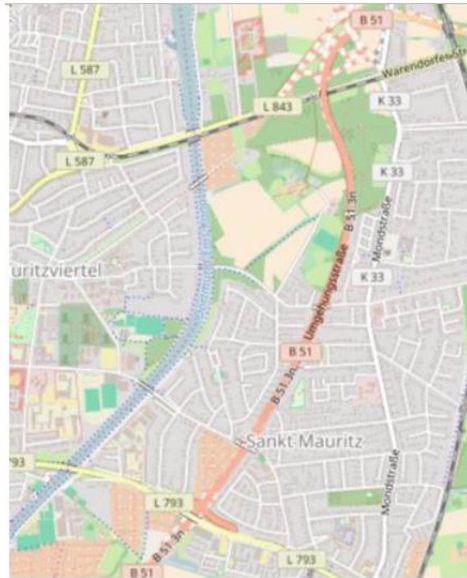
- CREAM WPS server
 - HAND (Height Above the Nearest Drainage) integrates additional ArcGIS functionalities
 - TauDEM (Terrain Analysis Using Digital Elevation Models) prepares a non time-dependent DTM from DEM
 - MiraMon cgi WPS derives flooded area for a selected date/time depending on water level measurements in stations by a SOS or pre-downloaded WaterML/CSV data
 - MiraMon model integrates the r.inund.fluv (GRASS) and spatialization procedure to the HAND method.



WaterInnEU - RIBASE

- Pub/Sub Component
 - RESTful component implemented by 52° North based on the OGC Pub/Sub standard
 - Publish: polygons (flooded Areas) are published from WPS to Pub/Sub component
 - Subscribe: Users can subscribe with a point-of-interest → notifications are sent, if the point intersects a flooded area

WaterInnEU - RIBASE



WPS Setup Capabilities Processes Execute

WPS Setup and Configuration

This is the basic configuration-tab. To use Web Processing functionalities you first have to choose an available service and also specify its version. With each re-selection (WPS or version), a GetCapabilities request will be executed. The capabilities response is then used to update the application and enable the remaining tabs. The tabs 'GetStatus' and 'GetResult' are permanently blocked for version 1.0.0.

Change Language

English

Please choose a WPS from the list:

Add WPS URL

Select the desired WPS service-version

WPS Capabilities

This area gives an overview of general information about the chosen Web Processing Service. An overview of all processes can be accessed from the 'Processes'-tab.

Identification

Provider

Operations

GetCapabilities

GET URL
http://www.ogc.uab.cat/cgi-bin/WaterInnEU/MiraMon.cgi?

POST URL
http://www.ogc.uab.cat/cgi-bin/WaterInnEU/MiraMon.cgi?

DescribeProcess

GET URL
http://www.ogc.uab.cat/cgi-bin/WaterInnEU/MiraMon.cgi?

POST URL
http://www.ogc.uab.cat/cgi-bin/WaterInnEU/MiraMon.cgi?

Execute

GET URL
http://www.ogc.uab.cat/cgi-bin/WaterInnEU/MiraMon.cgi?

POST URL
http://www.ogc.uab.cat/cgi-bin/WaterInnEU/MiraMon.cgi?

WPS Setup **Capabilities** Processes Execute

WPS Processes

This area provides you with information about the selected process, divided in 'common information' and the 'inputs' and 'outputs' of the process.

Available Processes

Flood areas generation from a DTM and from measurements and stations provided

General Information

Title

Flood areas generation from a DTM and from measurements and stations provided (Option 1)

Identifier

MiraMon.Inundac.Param01:1

Description

Flood areas generation at specified data from a DTM and from measurements and stations indicated by the user (Option 1). The point station measurements are spatialized through Thiessen polygons.

Process -Version

1

Available Output Transmission Modes

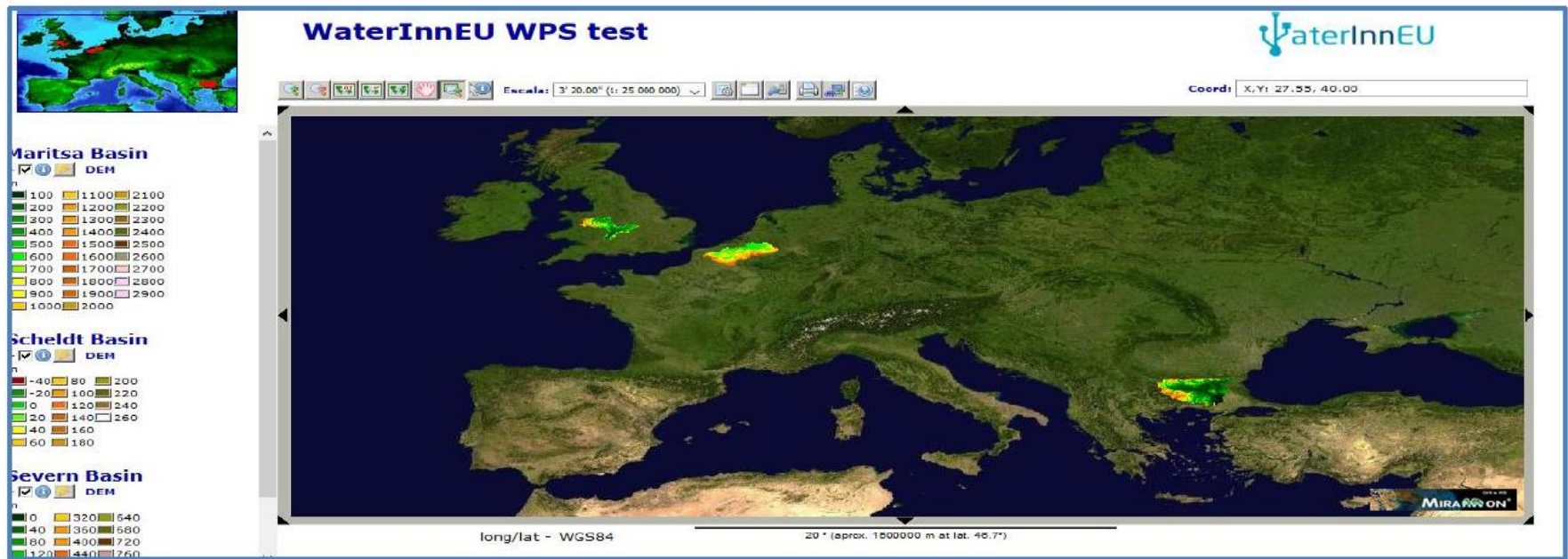
["value","reference"]

Available Job Execution Modes

["sync-execute","async-execute"]

Success! The DescribeProcess request for the currently selected process was successful.

WaterInnEU - RIBASE



Execute a process (WPS)

Operation to execute:
Flood areas generation from a DTM and from measurements and provided stations

Input parameters:
Option:
3

River Basin identifier:
Maritsa

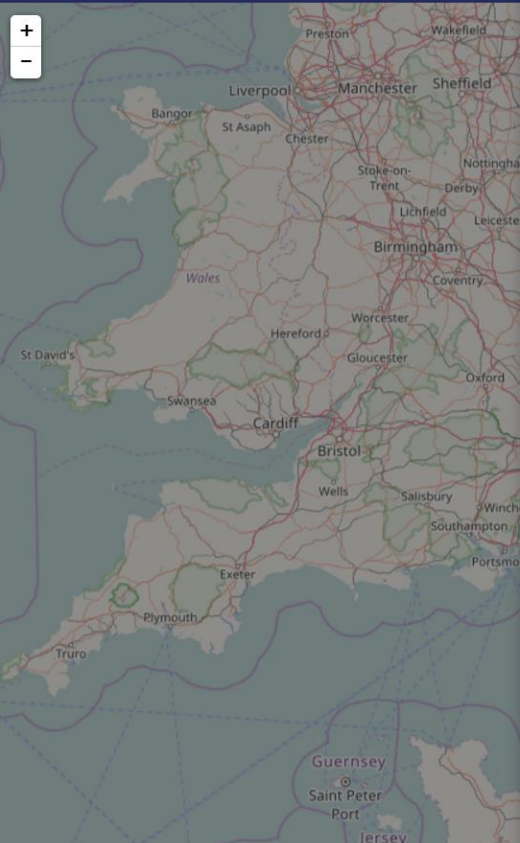
Date-time in ISO format:
2016-05-28T00:00:00.000

WaterInnEU - RIBASE

Eventing REST Api Client

+

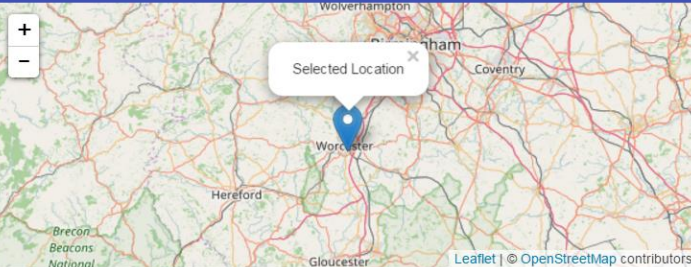
-



Add Subscription

+

-



Selected Location

Actual marker position (lat/lng): 52.17393169256849 / -2.208251953125

Subscription Label

Label

 Worcester Downton

End of Life

My date(dd/mm/yyyy)

 15/02/2017

 13:56

Delivery Methods

☐ Websocket delivery

☐ Email delivery

CANCEL

ADD

Leaflet | © OpenStreetMap contributors

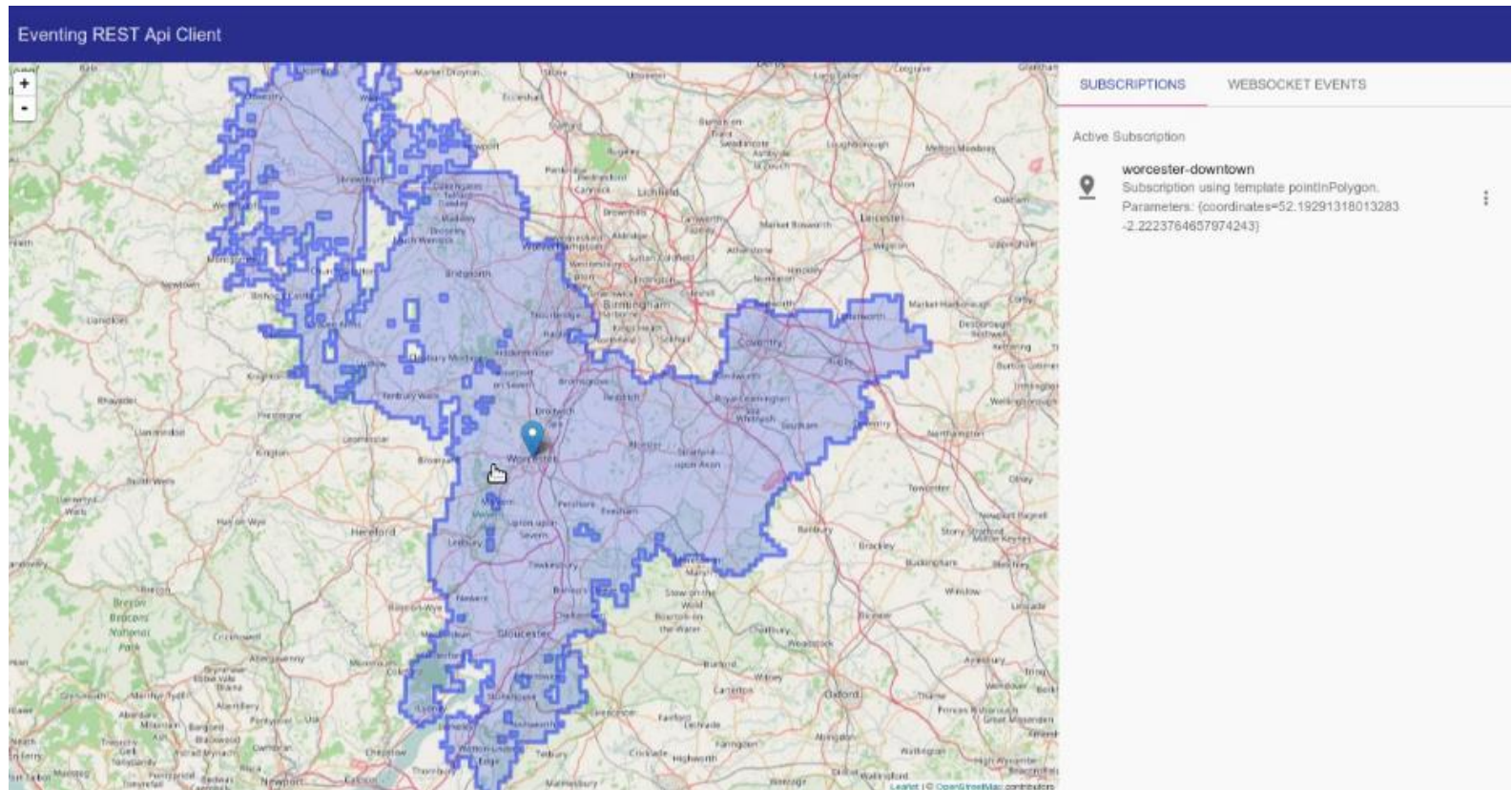
SUBSCRIPTIONS

WEBSOCKET EVENTS

Active Subscription



WaterInnEU - RIBASE



WaterInnEU - RIBASE

- The exchange of open hydrological information in standard formats and interoperable services works
- More interoperable data sources are desirable
- Data provision by SOS service in WaterML 2.0 is the recommended option for allowing to chain automatic procedures and applications
- Additional efforts are needed to support WPS but it offers advantages in terms of modularity, reusability, and interoperability
- The preparation of non-time dependent processing data is recommended for efficiency
- <http://ceur-ws.org/Vol-1762/Pesquer.pdf>

WaCoDiS

- Combination of heterogeneous data sources and existing interoperable web-based information systems
- Connection to the Copernicus infrastructure and the extension of INSPIRE-compliant Sensor Web technology to deal with big raster data
- Innovative analyses of high temporal resolution Sentinel-1 and Sentinel-2 Copernicus satellite data that contributes to the exploration of heavy rain effects on agricultural areas
- Integration of in-situ and satellite data into domain-oriented models to optimize the simulation of pollutant flows
- <https://wacodis.fbg-hsbo.de/>



WaCoDiS

Domain Applications



Water Industry
Associations



Suppliers



Agricultural
Sector



Public

Analytics Services
(Environmental Monitoring)

Business Models (material input etc.)

...

SDI

SDI

SDI
(Geodata and
Services)

Water Industry
Geodata Services

Water Industry
in-situ data
(Sensor Web)

Open API

Open/External
Measurement and
Forecast Data

Open API

Open remote
Sensing Data
(Copernicus)



WaCoDiS

- Current status: Architecture development
- Important challenges:
 - Integrating remote and in-situ measurements
 - Processing infrastructure
 - Event-based workflows

MuDaK-WRM

- Multidisciplinary Data Acquisition: The Key for a globally applicable Water Resource Management
- Main goals
 - Identification of all relevant parameters influencing the long-term behavior of a reservoir
 - Development of a minimum monitoring concept
 - Reduction of complexity and data demand of given model approaches
 - Development of a globally applicable tool for surface water resources

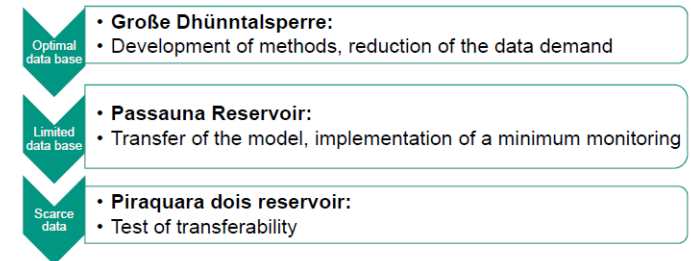
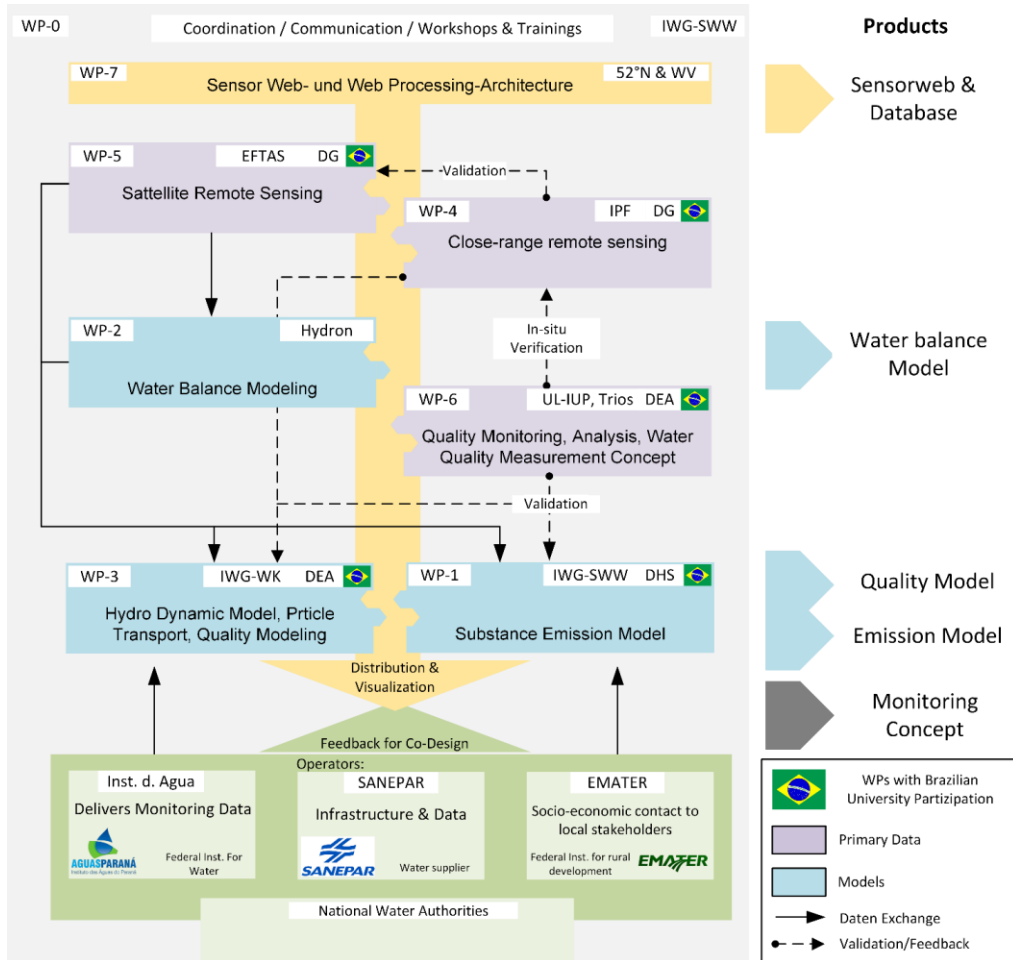


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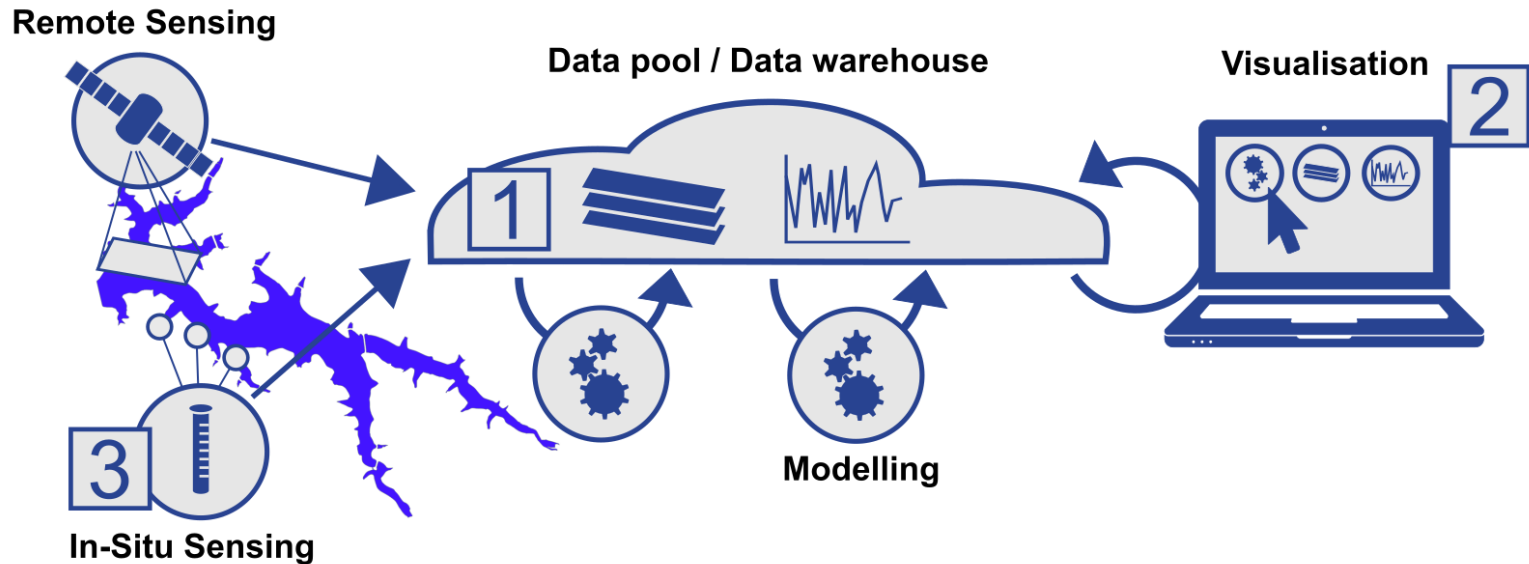


Federal Ministry
of Education
and Research

MuDaK-WRM



MuDaK-WRM

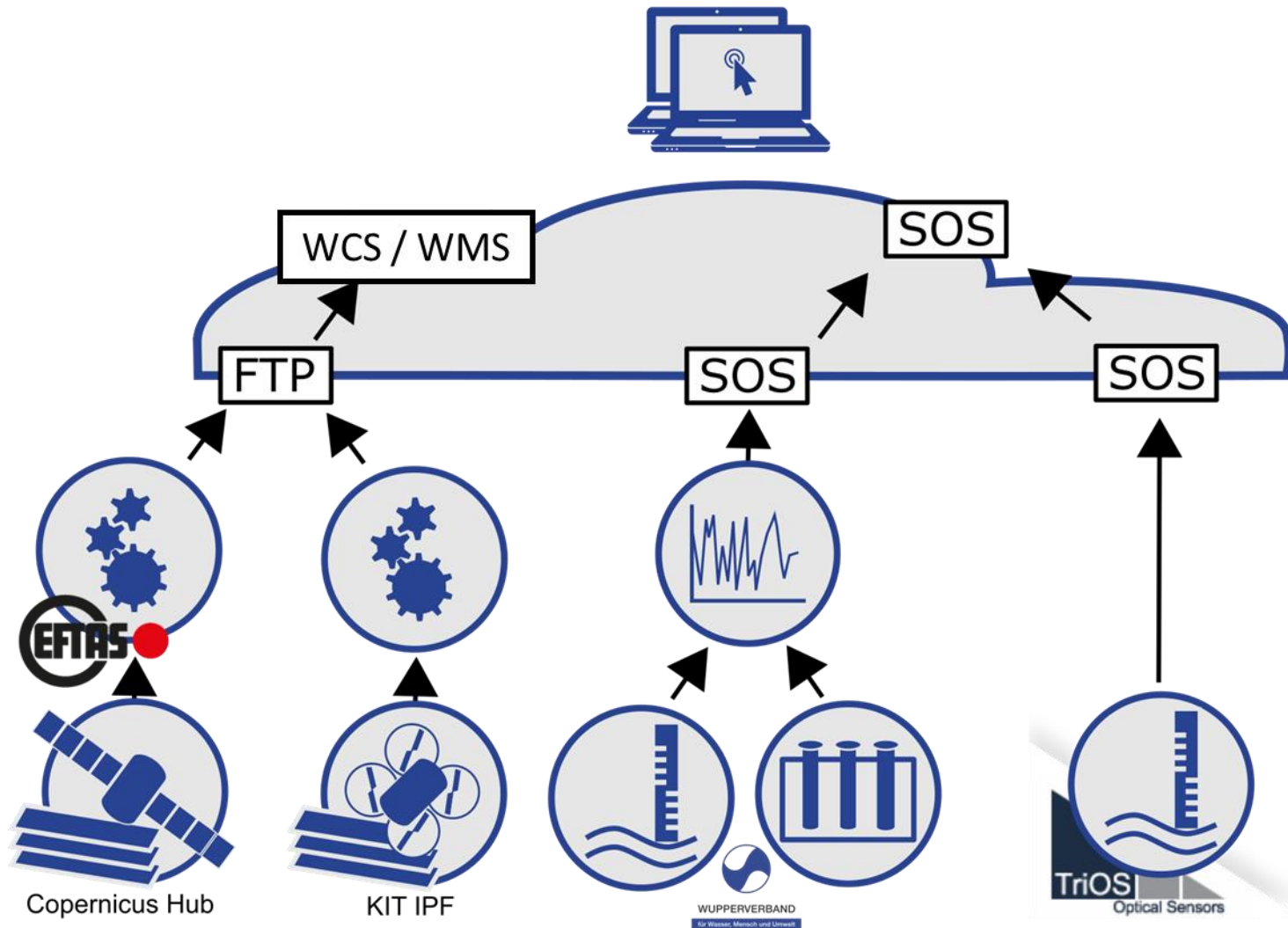


- 1** Build a central observation data exchange system
- 2** Visualise water quality data and campaigns
- 3** Integrate in-situ measurements with remote sensing data
- 4** Wrap models in inter-operable web processing services

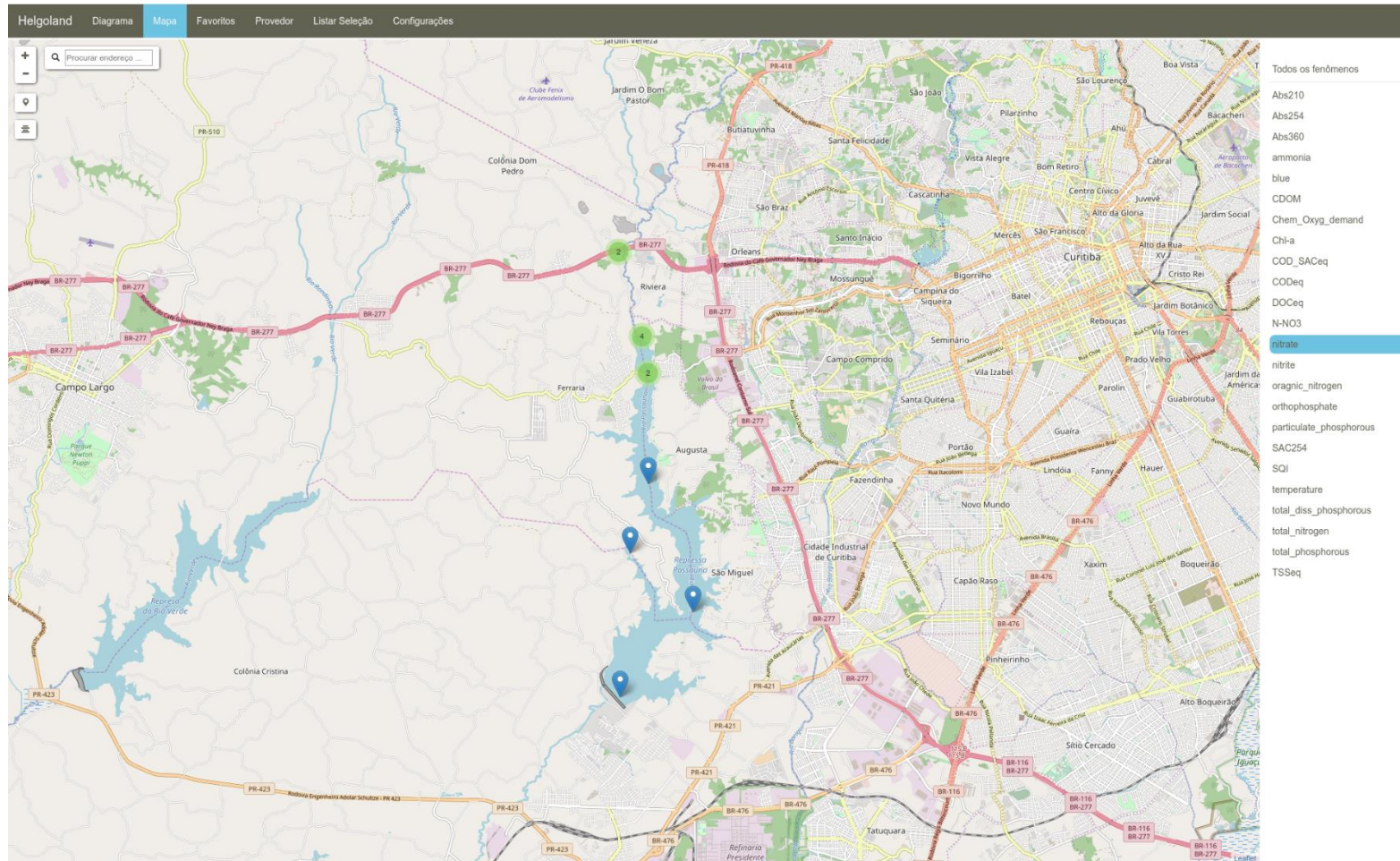
MuDaK-WRM

- Huge variety of heterogeneous data sets (~ 70)
- In-situ, UAV and satellite measurements
- Spatial:
 - Vector: point
 - Raster: 0.5 m – 10 m – 20 m
- Temporal:
 - Instant
 - Period: 30 sec – hourly – weekly – monthly – annual
- Various measurement units
- Typically of relevance for several partners

MuDaK-WRM



MuDaK-WRM



Thank you for your Attention!

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